

Pneumatic artificial muscles: A switching Model Predictive Control approach



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ABSTRACT

In this article, a switching Model Predictive Controller (sMPC) for a pneumatic artificial muscle (PAM) is presented. The control scheme is based on a switching PieceWise Affine (PWA) system model approximation that is able to capture the high nonlinearities of the PAM, while improving the overall model accuracy, and is composed of: (a) a feed-forward term regulating control input at specific reference set-points, and (b) a switching Model Predictive Controller handling any deviations from the system's equilibrium points. Extended experimental studies are being presented that prove the overall scheme's efficiency.

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1. Introduction

Pneumatic artificial muscle (PAM) (Pierce, 1940; Warszawski, 1967), also known as the pneumatic muscle actuator or fluidic muscle (Gaylord, 1958; Yarlott, 1972) is a tube-like actuator that, when pressurised, decreases in actuating length. The McKibben artificial muscle (Chou & Hannaford, 1996; Schulte, 1961), is the most utilized type of PAMs, invented in 1950s by Joseph L. McKibben and utilized as an orthotic appliance for polio patients (Nickel, Perry, & Garrett, 1963).

PAM typically consists of a rubber tube wrapped inside double helix synthetic fiber netting at predetermined angle. Protective rubber coating surrounds the fiber wrapping and appropriate metal fittings are attached at each end. The fluidic muscle—one of the most utilized commercially available muscle-type actuators in the automation industry, like the one presented in Fig. 1, differs from the classic McKibben PAM as it utilizes a combined integration of the inner tube and external shell to a single aramid-neoprene mesh.

PAM possesses similar properties with those of the organic muscle, combined with several advantages as the ability to provide high power outputs, with relatively light weights and inherent compliance. These characteristics are turning the PAM into a promising actuator choice in Anthropomatic technologies, while a variety of such applications have been appeared in the literature (Tsagarakis & Caldwell, 2002; Herr & Kornbluh, 2004). Moreover, PAM meets the

need for safety, simplicity and lightness that human–robot interaction requires and justifies its expanding utilization in medical and biorobotic applications (Andrikopoulos, Nikolakopoulos, & Manesis, 2011).

PAMs' highly nonlinear nature causes difficulties in modeling their behavior and designing controllers for high-performance positioning systems, while in most industrial PAM-actuated operations (e.g. aligning, pressing, drilling, gripping, clamping, handling, transporting, or assembling) fast and accurate position control is of utmost importance. Thus, PAMs present a challenging modeling and control problem. In addition, a bibliographical research on PAM related applications, shows the significance of researching an optimal solution to the positioning control problem (Andrikopoulos et al., 2011).

Some of the most significant control design approaches for PAM have been developed during the past fifteen years. Those control schemes have included a neural network based training algorithm (Hesselroth, Sarkar, Van der Smagt, & Schulten, 1994), adaptive control (Medrano-Cerda, Bowler, & Caldwell, 1995), variable structure control (Repperger, Johnson, & Phillips, 1998), gain scheduling model-based control (Repperger, Phillips, & Krier, 1999), model-based nonlinear control (Krichel, Sawodny, & Hildebrandt, 2010), cascaded control (Hildebrandt, Sawodny, Neumann, & Hartmann, 2005), fuzzy control (Chang & Lilly, 2003; Balasubramanian & Rattan, 2003), nonlinear optimal predictive control (Reynolds, Repperger, Phillips, & Bandry, 2003), direct continuous-time adaptive control (Lilly, 2003a, 2003b), PID control (Chan, Lilly, Repperger, & Berlin, 2003; Ganguly, Garg, Pasricha, & Dwivedy, 2012), neural network nonlinear PID control (Ahn & Thanh, 2005), sliding-mode control (Lilly, 2003a, 2003b; Aschemann & Schindele, 2008), direct/indirect adaptive-robust control (Zhu, 2009), chattering-free robust variable structure

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Fig. 1. A pneumatic artificial muscle (PAM) during relaxation.

control (Choi & Lee, 2010), compliance control (Choi, 2011), hybrid distributed macro-mini control (Shin, Sardellitti, & Khatib, 2008), proxy sliding mode control (Damme et al., 2009) and multi-parametric constrained optimal control (Andrikopoulos, Nikolakopoulos, Arvanitakis, & Manesis, 2012a, 2012b).

Nowadays, the state of the art research is focusing in both seeking novel applications based on PAM actuation, while increasing the general efficiency of PAMs as linear actuators, where fast and very accurate positioning systems are needed (Tsagarakis & Caldwell, 2002). In the presented literature survey, regarding control schemes for PAMs, the utilized models are simple, since the effort has been dedicated to the application side of view, while the control schemes are fundamental and none of them has the merit to take under consideration physical and mechanical constraints acting on the system states. In the presented research effort a novel modeling procedure will be presented in accordance to a properly tuned control framework being able to provide an improved modeling and control (positioning) accuracy.

In this approach, the PAM is considered as a basic pneumatic actuator and issues regarding the integration of the actuator in more complex systems are out of the scope of the presented research. However, the scientific results that will be presented and have been experimentally evaluated, from an engineering point of view and without losing generalization, can form the basis for increasing the overall performance and the accuracy of more complicated PAM devices, e.g. exoskeletons.

The proposed research has been inspired by the increased trend in installing and utilizing PAMs in industrial applications. In such cases, the PAM is being treated as a linear actuator for accomplishing various tasks such as pressing, drilling, gripping, transporting or assembling operations. All these applications base their proper operation on the effectiveness of the positioning control problem, while after a bibliographical search in the related literature it is obvious that most of the attention in the area of PAMs have been their utilization on novel designs and the full utilization of their physical properties, which resulted in a corresponding less attention in control approaches that could achieve very fast and extremely accurate positioning. For satisfying this need and after reviewing the existing control approaches, the switching MPC scheme has been selected as the solution that would support: (a) PWA modeling of the highly PAM nonlinearities, (b) switching system representations to account for both the modeling approach and the hysteretic phenomena during inflation and deflation, and (c) the ability to handle physical and operational constraints, as these constraints could be embedded during the cost optimization process.

The novelty of the proposed modeling and control scheme is dual. First, a switching PieceWise Affine (PWA) modeling approach (Andrikopoulos et al., 2012a, 2012b) is being ameliorated to account for the alteration of PAM dynamics during inflation and deflation operating states due to hysteretic phenomena. This

improvement results in an increase in the accuracy of the model under different set-points, while maintaining an accessible mathematical description (Sontag, 1981). Second, a model predictive control (MPC) scheme is being experimentally applied and validated to the case of a PAM.

Moreover, one of the novelties of the presented work is that instead of developing a control scheme based on PAM's non-linear model, the system dynamics are approached by a multiple model representation. The adopted PWA description can mathematically represent systems that present different dynamics depending on the specific point in the state-input space under consideration.

In general, MPC (Richalet, Rault, Testud, & Papon, 1978) is a form of on-line control design method that solves the finite-horizon optimal control problem for the current system state, rather than computing the feedback law in an off-line matter (Mayne, Rawlings, Rao, & Sckaert, 2000). Among its benefits are the relative ease of tuning, the ability to control multivariable plants and take account of actuator and physical limitations by handling soft and hard system constraints, as well as the advantage of tracking pre-scheduled reference signals (Maciejowski, 2001).

At this point it should be clarified that the proposed switching MPC is relying on a PWA system modeling representation and a continuous state switching mechanism, while the optimization cost will be solved by Quadratic Programming. The presented approach differs from the mixed logical dynamical system (MLD) modeling procedure, since the system model representation will not be extended to include integer parameters as states and thus the optimization cost will utilize the mixed integer quadratic programming (MIQP) approach (Heemels, De Schutter, & Bemporad, 2001; Bemporad & Morari, 1999).

In this article, the linearization of the PAM's linear motion dynamics at several operating points results in a family of PWA linear systems, where each system is valid for a subset of the PAM's motion interval. The proposed switching Model Predictive Control (sMPC) scheme results in an optimal control of each operating region and, as the system switches from one region to its neighboring ones, ensures the control effort's smooth transition.

The article is structured as it follows. In Section 2, a switching PWA system representation, derived from the fundamental modeling for static and dynamic PAM performance, will be presented. In Section 3, a controller composed of: (a) a feed-forward term regulating control input at specific set-points, and (b) a switching model predictive (sMP) controller handling any deviations from the system's equilibrium points, is being synthesized. In Section 4, multiple experimental studies are being presented that prove the efficacy of the sMPC in the contraction positioning problem of a PAM, while conclusions will be drawn in Section 5. In Appendix A, a list of variables and symbols utilized in this paper is being provided for clarification purposes.

2. Pneumatic artificial muscle modeling

In this section, an analysis of the most commonly utilized modeling approaches for describing the static and dynamic performance of PAMs, will be presented, followed by the proposed novel switching PWA system representation with augmented system dynamics to account for the design and utilization of the sMPC scheme.

2.1. Force model

In Fig. 2, the basic geometrical characteristics of the PAM during the relaxed and inflated state are being presented, while the utilized variables are being defined as: l is the tube length, r is

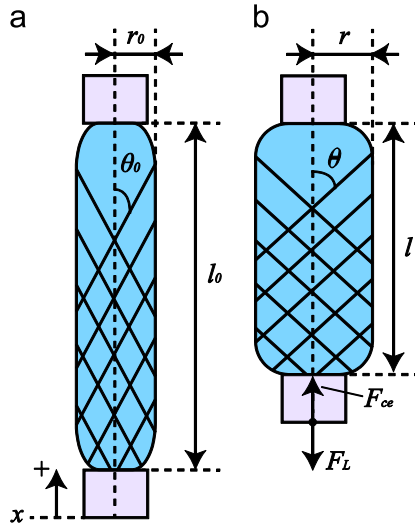


Fig. 2. Simplified geometrical model of PAM during the (a) relaxed and (b) inflated state under load.

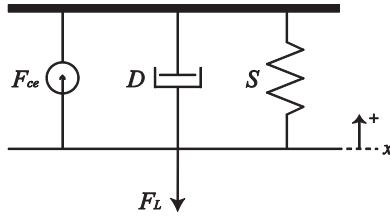


Fig. 3. Three-element dynamic model of PAM.

the bladder radius, and θ is the braid angle, which is the angle between the helical fiber element length and the longitudinal axis of the actuator (Schulte, 1961). As depicted in (1), the contractile force F_{ce} is described as linearly proportional to the relative pressure P , i.e. the absolute internal gas pressure minus the environment pressure, and a monotonic function of braid angle θ (Tondur & Lopez, 2000).

$$F_{ce} = \pi r_0^2 P [a(1 - \varepsilon)^2 - b], \quad (1)$$

where $a = (3 / \tan^2 \theta_0)$, $b = (1 / \sin^2 \theta_0)$, $\varepsilon = (l_0 - l / l_0)$ and r_0 , l_0 , θ_0 are the initial radius, the initial nominal length, and the initial braid angle, respectively. The inflated state is being accompanied by an external force F_L , which plays the role of an external load and has a fundamental effect in the following analysis of the PAM dynamics.

2.2. PAM dynamics

During expansion, PAM experiences viscoelastic resistance which can be modeled as a damping and spring element. Thus, PAM is being considered as a parallel pattern that consists of a spring element S , a damping element D and a contractile element F_{ce} as presented in Fig. 3 (Reynolds et al., 2003). This model corresponds to a PAM placed on the vertical position with one end fixed and an external load F_L attached to the other end. The differential equation that describes the overall system can be formulated as:

$$M\ddot{x} + D\dot{x} + Sx = F_{ce} - F_L, \quad (2)$$

where $x \in \mathfrak{R}^+$ is the displacement of the PAM, M is the mass of the muscle, F_L is the external load, S is the spring constant, D is the damping coefficient and F_{ce} is the contractile muscle force as presented in (1).

2.3. PieceWise Affine system model approximation

In many application areas and specifically for highly non-linear systems, instead of developing a control scheme based on the non-linear model it is straightforward and simpler to approach the non-linear system dynamics by a multiple model representation. In other words, a PWA description is suitable for a system that evolves according to different dynamics depending on the specific point in the state-input space under consideration. In the related literature, PWA systems are the object of extensive research as: (i) they constitute an immediate extension of linear systems capable of modeling nonlinear/non-smooth phenomena with an arbitrary degree of precision and (ii) they allow to consider fundamental hybrid features such as linear-threshold events and mode switching.

In the examined case, the linearization of the PAM's linear motion dynamics at several operating points results in a family of PWA linear systems, where each system is valid for a subset of the PAM's motion interval. In the sequel, a state-space representation of the PAM's motion dynamics is being formulated, accompanied by a PWA segmentation that improves the static results by increasing the accuracy of the model under different set-points, while maintaining an accessible mathematical description (Sontag, 1981).

The operating points x_j^{op} , with $j = \{1, \dots, E\}$ and $E \in \mathbb{Z}^+$, depend on the applied nominal pressures P_j^{op} , under the assumption of a constant load. Eq. (2) for $\ddot{x} = \dot{x} = 0$ yields:

$$P_j^{op} = \frac{Sx_j^{op} + F_L}{\pi r_0^2 [a(1 - (x_j^{op} / l_0))^2 - b]}. \quad (3)$$

The linearized equations of motion for the PAM, around multiple equilibrium points j can be formulated by considering small perturbations around the variables x and P as: $x = x_j^{op} + \delta x$ and $P = P_j^{op} + \delta P$, respectively.

By applying this linearization approach in (2), it yields:

$$M\delta\ddot{x} + D\delta\dot{x} + Sx_j^{op} + S\delta x = F_{ce}(P_j^{op} + \delta P, x_j^{op} + \delta x) - F_L. \quad (4)$$

Considering the Taylor expansion of (1) around the operating points and by utilizing the first two order terms:

$$\begin{aligned} F_{ce}(P_j^{op} + \delta P, x_j^{op} + \delta x) = & \pi r_0^2 P_j^{op} \left[a \left(1 - \frac{x_j^{op}}{l_0} \right)^2 - b \right] \\ & + \pi r_0^2 \left[a \left(1 - \frac{x_j^{op}}{l_0} \right)^2 - b \right] \delta P \\ & - \pi r_0^2 P_j^{op} \left[\frac{2a}{l_0} \left(1 - \frac{x_j^{op}}{l_0} \right) - b \right] \delta x, \end{aligned} \quad (5)$$

and by substituting (3)–(5) yields:

$$\begin{aligned} F_{ce}(P_j^{op} + \delta P, x_j^{op} + \delta x) = & (Sx_j^{op} + F_L) + \pi r_0^2 \left[a \left(1 - \frac{x_j^{op}}{l_0} \right)^2 - b \right] \delta P \\ & - \frac{(Sx_j^{op} + F_L)[(2a/l_0)(1 - (x_j^{op}/l_0))] }{[a(1 - (x_j^{op}/l_0))^2 - b]} \delta x. \end{aligned} \quad (6)$$

Inserting the expression from (6) to (4) is being obtained:

$$M\delta\ddot{x} + D\delta\dot{x} + Q\delta x = R\delta P, \quad (7)$$

where $Q = S + ((Sx_j^{op} + F_L)/[a(1 - (x_j^{op}/l_0))^2 - b])[2a/l_0(1 - (x_j^{op}/l_0))]$, which by taking (3) into account becomes $Q = S + P_j^{op} \pi r_0^2 [2a/l_0(1 - (x_j^{op}/l_0))]$, and $R = \pi r_0^2 [a(1 - (x_j^{op}/l_0))^2 - b]$.

The addition of term $M\ddot{x}_j^{op} + D\dot{x}_j^{op} + Qx_j^{op}$ on both sides of (7) yields:

$$M\ddot{x} + D\dot{x} + Qx = Qx_j^{op} + R\delta P. \quad (8)$$

Finally, Eq. (8) can be re-written as:

$$\ddot{x} + \frac{D_j}{M} \dot{x} + \frac{Q_j}{M} x = \frac{Q_j}{M} x_j^{op} + \frac{R_j}{M} \delta P, \quad (9)$$

where the j th subscript has been utilized to denote the dependence of the previous variables on the j -selected operating point.

The equivalent multiple state space models, accounting for small perturbations around the operating point x_j^{op} , are described as:

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{Q_j}{M} & -\frac{D_j}{M} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{R_j}{M} \end{bmatrix} \delta P + \begin{bmatrix} 0 \\ \frac{Q_j}{M} x_j^{op} \end{bmatrix}, \quad (10)$$

or in an equivalent compact state space representation as:

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}_j \mathbf{x} + \mathbf{B}_j \delta P + \mathbf{f}_j, \\ \mathbf{y} &= \mathbf{C}_j \mathbf{x} + \mathbf{D}_j \delta P, \end{aligned} \quad (11)$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \in X \subseteq \mathbb{R}^2, \mathbf{y} = \mathbf{x}, \text{ and } \delta P \in P \subseteq \mathbb{R},$$

with the sets X and P to specify state and input operating regions that contain the operating points in their interior, and

$$\begin{aligned} \mathbf{A}_j &= \begin{bmatrix} 0 & 1 \\ -\frac{S_j}{M} - \frac{2\pi r_0^2 a}{M l_0} P_j^{op} \left(1 - \frac{x_j^{op}}{l_0}\right) & -\frac{D_j}{M} \end{bmatrix}, \\ \mathbf{B}_j &= \begin{bmatrix} 0 \\ \frac{\pi r_0^2}{M} \left[a \left(1 - \frac{x_j^{op}}{l_0}\right)^2 - b \right] \end{bmatrix}, \\ \mathbf{C}_j &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{D}_j = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \end{aligned}$$

with the affine term denoted as

$$\mathbf{f}_j = \begin{bmatrix} 0 \\ \left[\frac{S_j}{M} + \frac{2\pi r_0^2 a}{M l_0} P_j^{op} \left(1 - \frac{x_j^{op}}{l_0}\right) \right] x_j^{op} \end{bmatrix}.$$

The system described in (10) is switched, as j belongs to a finite set of indexes and E denotes the number of switched systems, with the state x to act as the switching rule. If Σ is the polytope that contains all the switching systems, defined by the switching vertices $[\mathbf{A}_j, \mathbf{B}_j]$, it can be notated as:

$$\text{Co}\{[\mathbf{A}_1, \mathbf{B}_1], \dots, [\mathbf{A}_E, \mathbf{B}_E]\}, \quad (12)$$

where the notation Co denotes the convex hull of the set, and E is the maximum number of allowed switches.

These linear time-invariant state space models can be transformed into their discrete equivalents under the assumption of a sampling process with sampling period T_s . The discrete models can be cast in a compact form as a set of PWA subsystems:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}_j^* \mathbf{x}(k) + \mathbf{B}_j^* \delta P(k) + \mathbf{f}_j^*, \\ \mathbf{y}(k) &= \mathbf{C}_j^* \mathbf{x}(k) + \mathbf{D}_j^* \delta P(k), \end{aligned} \quad (13)$$

where $k \in \mathbb{Z}^+$, $[\mathbf{x}(k) \ \delta P(k)]^T \in \Omega_i$ with Ω_i a polyhedral partition of the state-input space over which the different dynamics are active and the upper script * notation denotes the discrete equivalents of the matrices in (11).

2.4. Augmented system dynamics for sMPC synthesis

For the formulation of the sMPC-controller, (13) is being modified to include the affine term in the state vector as it follows:

$$\begin{aligned} \mathbf{x}^{au}(k+1) &= \mathbf{A}_{j,au}^* \mathbf{x}^{au}(k) + \mathbf{B}_{j,au}^* \delta P(k), \\ \mathbf{y}^{au}(k) &= \mathbf{C}_{j,au}^* \mathbf{x}^{au}(k) + \mathbf{D}_{j,au}^* \delta P(k), \end{aligned} \quad (14)$$

where

$$\begin{aligned} \mathbf{A}_{j,au}^* &= \begin{bmatrix} \mathbf{A}_j & \mathbf{V} \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}, \mathbf{B}_{j,au}^* = \begin{bmatrix} \mathbf{B}_j \\ 0 \end{bmatrix}, \mathbf{C}_{j,au}^* = \begin{bmatrix} \mathbf{C}_j & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & 1 \end{bmatrix}, \\ \mathbf{D}_{j,au}^* &= \begin{bmatrix} \mathbf{D}_j \\ 0 \end{bmatrix}, \mathbf{V} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{x}^{au}(k) = \begin{bmatrix} \mathbf{x}(k) \\ f_j^{au} \end{bmatrix} \in X^{au} \subseteq \mathbb{R}^3, \text{ and} \\ f_j^{au} &= \left[\frac{S_j}{M} + \frac{2\pi r_0^2 a}{M l_0} P_j^{op} \left(1 - \frac{x_j^{op}}{l_0}\right) \right] x_j^{op}. \end{aligned}$$

The augmented state space representation (14) of PAM motion dynamics will be utilized in the proposed novel sMPC synthesis, as described in Section 3.

3. Switching Model Predictive Controller synthesis

3.1. Discrete system model formulation

Consider the discrete time constrained augmented PieceWise-Affine system described by (14). The states of this system can be directly measured, while for calculating the future predictions of the system, based on $y(k)$ and $u(k)$, the following recursions should be incorporated based on (14).

$$\begin{aligned} \mathbf{x}^{au}(k+i+1|k) &= \mathbf{A}_{j,au}^* \mathbf{x}^{au}(k+i|k) + \mathbf{B}_{j,au}^* \delta P(k+i), \\ \mathbf{y}^{au}(k+i|k) &= \mathbf{C}_{j,au}^* \mathbf{x}^{au}(k+i|k) + \mathbf{D}_{j,au}^* \delta P(k+i), \end{aligned} \quad (15)$$

For clarity of the following presentation and for simplifying the mathematical notations, in the sequel the control variable $\delta P(k)$ will be denoted as $u(k)$. Eq. (15) state that $y(j+i|k)$ is a linear function of x and u . Moreover the control increments are being defined as $\Delta u(k) = u(k) - u(k-1)$. In the presenting notation, $k+i|k$ denotes the predicted value for time $k+i$ based on information available at the k -time instant.

Thus, the j dynamics represented by the tuple $[\mathbf{A}_{j,au}^*, \mathbf{B}_{j,au}^*, \mathbf{C}_{j,au}^*, \mathbf{D}_{j,au}^*, f_j^{au}]$, will be active in the part of state-input space, which satisfies the posed constraints as it will be presented in the sequel. The same equation partitions the $(P \times X^{au}) \subseteq \mathbb{R}^4$ space into a set of polyhedral Ω . Based on this partitioning and the current value of the state vector, the corresponding dynamics are initially being selected, while in the sequel these dynamics are being utilized by the MPC scheme to produce the future values for the system's states and to solves the corresponding optimization cost (calculate an optimal control effort) in an online manner as it will be defined in the sequel.

3.2. sMPC synthesis

The design of the proposed sMPC-scheme is based on: (a) a feed-forward controller that utilizes (3) to generate the control effort P_0 based on the desired position x_{ref} , and (b) a sMPC-controller applied on the multiple PWA PAM system representations, which generates the additional stabilizing $u(k)$ control effort by performing online minimization of an optimal control cost.

The proposed dual scheme, as presented in Fig. 4, takes into account the alteration of PAM dynamics during inflation and deflation operating states due to hysteretic phenomena and is designed to compensate for the intense nonlinear characteristics and thus achieving high accuracy control.

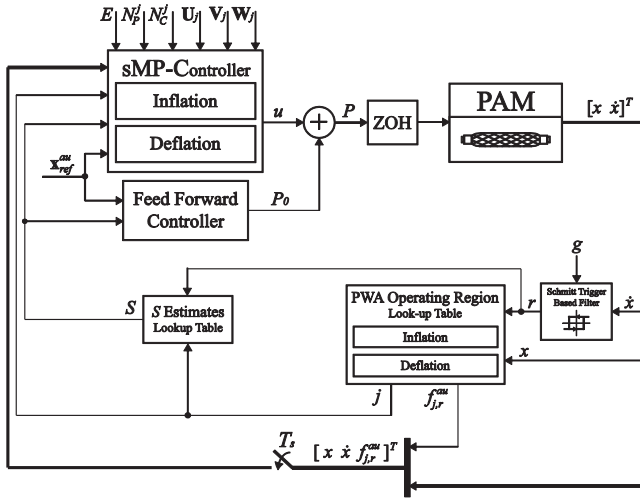


Fig. 4. Feed-forward and switching Model Predictive Control scheme.

A switching signal r is being utilized for activating the appropriate sMP-controller, depending on the operating state that the reference signal imposes. This signal is being calculated by extracting the sign of the state velocity $\dot{x}$ from the state feedback. In order to avoid uncontrollable switching, due to minor measurement inaccuracies of the laser distance sensor, a Schmitt trigger-based filter (Schmitt, 1938) is also imposed on the signal r . The aforementioned switch signal is being denoted as

$$r(k) = \begin{cases} 1 & \text{for } \dot{x}(k) \geq g \\ r(k-1) & \text{for } |\dot{x}(k)| < g \\ -1 & \text{for } \dot{x}(k) \leq -g \end{cases} \quad (16)$$

where $g = 0.03$ m/s is an empirical fine-tuned constant, derived from extended experimental trials by taking into account the measurement accuracy of the laser distance sensor.

The parameter g is being experimentally defined in order to avoid unnecessary switching between inflation and deflation states caused by measurement fluctuations, and is associated the measurement accuracy of the distance sensor. This phenomenon is mainly observed during steady states or slow reference signals where the position is governed by small changes due to the sensor's limited accuracy, causing unnecessary switching due to false readings. The easiest way to surpass that effect was to observe such cases and keep the switch signal r unaltered when the velocity is bounded by a minimum velocity marker denoted here as g . Experimental studies showed that $g = 0.03$ m/s. Smaller values of g resulted in the aforementioned unnecessary switching, whereas larger values reduce the controller's responsiveness to operating state transitions.

Furthermore, the j th PWA operating region, which corresponds to the current value of the x -state and the switching signal r , is extracted via a generated lookup-table, denoted in Fig. 4 as "PWA Operating Region Look-up Table". The spring coefficient S calculation, corresponding to the current value of r and the active PWA subsystem j , is also extracted via a lookup-table and utilized for the calculation of the feedforward control effort P_0 via (3). The process for calculating the spring coefficient will be analytically presented in Section 4.2, while here a brief overview has been provided to explain the whole operation of the proposed control scheme.

The resulting sMP-controller consists of $2E$ MP-controllers, each one set E has been fine-tuned to best control the corresponding PWA system representation and operating states. In a generic framework, the j th MP-controller performs on-line computation of

a finite-time optimal control sequence by utilizing the current system state vector value as an initial optimization state.

3.3. sMPC cost objective and constraints

The controller's objective is to optimize the quadratic cost in (17). In particular, the MP-controller that corresponds to the j th PWA system representation of the PAM's displacement at discrete time k is obtained by minimizing the following quadratic program (QP) optimization cost with respect to the small control moves Δu and the relaxing slack variable ε :

$$J_j(\Delta u, \varepsilon) = \sum_{i=0}^{N_p-1} \left\{ \sum_{l=1}^{N_p} |W_{i+1,l}^j [y_l^{au}(k+i+1|k) - y_{ref,l}^{au}(k+i+1)]|^2 + \sum_{l=1}^{N_c} |V_{i,l}^j \Delta u_l(k+i|k)|^2 + \sum_{l=1}^{N_c} |U_{i,l}^j (u_l(k+i|k) - u_{target,l}(k+i))|^2 \right\} + \rho_\varepsilon \varepsilon^2, \quad (17)$$

where y_l^{au} and $y_{ref,l}^{au}$ are the output and reference signal elements of the augmented controlled system; $u_{target,l}(k+i)$ is an input set-point; ρ_ε is a penalty weight on the slack variable ε ; $U_{i,l}^j$, $V_{i,l}^j$, and $W_{i+1,l}^j$ are the non-negative penalty weights on the input u , input increments Δu , and the outputs y of the j th MP-controller, respectively, which form the diagonal penalty matrices U_j , V_j , and W_j . The subscript $(\cdot)_l$ denotes the l th component of a vector; $k+i|k$ denotes the predicted value for time $k+i$ based on information available at the k -time instant; N_p^j and N_c^j are the prediction and control horizon of the j th MP-controller, respectively. Moreover, the sequence of control effort increments $\{\Delta u(k|k), \dots, \Delta u(m-1+k|k)\}$ is being defined as $u(k) = u(k-1) + \Delta u(k|k)^*$, where $\Delta u(k|k)^*$ is the first element of the resulting optimal sequence.

One of the most important benefits of the proposed switching MPC scheme is its ability to take under consideration constraints on the states and the control actions. More specifically, the MPC has the merit to solve the optimization cost in (17) based on the following mathematically posed constraints, while different segments of the polyhedral partitions are defined using constraints on state and input variables as:

$$\begin{bmatrix} u_{l, \min}(i) - \varepsilon U_{l, \min}^j(i) \\ \Delta u_{l, \min}(i) - \varepsilon V_{l, \min}^j(i) \\ y_{l, \min}(i) - \varepsilon W_{l, \min}^j(i) \end{bmatrix} \leq \begin{bmatrix} u_l(k+i|k) \\ \Delta u_l(k+i|k) \\ y_l(k+i+1|k) \end{bmatrix} \leq \begin{bmatrix} u_{l, \max}(i) + \varepsilon U_{l, \max}^j(i) \\ \Delta u_{l, \max}(i) + \varepsilon V_{l, \max}^j(i) \\ y_{l, \max}(i) + \varepsilon W_{l, \max}^j(i) \end{bmatrix}, \quad \Delta u(k+h|k) = 0, \quad i = 0, \dots, N_p^j, \quad h = N_c^j, \dots, N_p^j - 1, \quad \varepsilon \geq 0, \quad (18)$$

While only the j th active MP-controller solves its QP optimization problem, all MP-controllers update their internal state observers, which ensures the control effort's smooth transition as the PAM system switches from one operating region to its neighboring ones.

In addition, the appropriate affine constant term f_j^{au} , which corresponds to the j th active augmented PWA system representation, is extracted by the x state operating region look-up table and inserted into the augmented state vector feedback. Thus, the affine state f_j^{au} is properly updated every time the system switches from one PWA subsystem to a neighboring one.

4. Experimental studies

4.1. Experimental setup

The wide utilization in automated industry, the commercial availability and the high nonlinear phenomena during operation, made the fluidic PAM type the most suitable choice for testing the control scheme's efficacy. Specifically, the experimental studies were carried on a Festo MXAM-40-AA-DMSP-40-305N-AM-CM fluidic muscle with 40 mm of internal diameter, 305 mm of nominal length, 840 g of mass, operating pressure range between 0 and 6 bar and allowable displacement of $x \in [0, 91.1]$ mm. As shown in Fig. 5, the PAM is on the vertical position with no additional load attached and has its upper end clumped.

In addition, a Festo VPPM-6F-L-1-F-0L6H-V1N proportional pressure regulator was used to regulate the supply and pressure of the compressed air supplied into the PAM. The pressure of the air supplied into the PAM is measured via the internal sensor of the proportional pressure regulator. A Festo SOEL-RTD-Q50-PP-S-7L laser distance sensor is being utilized to measure the under study PAM's displacement in the vertical axis. The motion of the PAM and the resulted displacement from its initial position is being measured via the aforementioned laser distance sensor and the use of a reflective surface that is properly adjusted to the lower cap of the muscle. Moreover, the pneumatic network utilized throughout the experimental studies is implemented via a Balma compressor with 2Hp of power and a maximum pressure of 11 bar, as well as a Combi compressor with 2Hp of power and a maximum pressure of 10 bar. The outputs of the two aforementioned compressors are connected in parallel, thus, supplying with pressurized air while avoiding any undesired pressure fluctuations. In the experiments that will be presented in the sequel, both compressor systems were operated simultaneously and in full load configuration, thus, minimizing the risk of pulsation or switching effects during experimentation.

The control of the setup operation, as well as the data acquisition, was achieved by utilizing a National Instruments USB-6215

Data Acquisition Card while the setup's programming is carried out in National Instruments LabVIEW, and Mathworks Matlab software.

4.2. Piecewise model parameter estimation

The operating regions' boundary elements are selected as the steady state values of the PAM system open loop inflation response for pressures from 0 to 600 kPa, in intervals of 100 kPa; whereas their respective linearization points are selected as the steady state values of the PAM system open loop response for the middle-point pressures P^{pp} from 50 to 550 kPa, with 100 kPa intervals. The same boundary elements were imposed on the deflation responses from the initial pressure of 600 kPa.

For the optimal estimation of the damping D and spring S coefficients in (2), extensive contraction studies were conducted. Considering the previous studies in (Reynolds et al., 2003) and (Andrikopoulos et al., 2012a, 2012b), at a steady state L both damping and spring coefficients are polynomial expressions of pressure P and can be formulated as:

$$D(P) = \sum_{i=0}^n d_i P^i, \quad (19)$$

$$S(P) = \sum_{i=0}^n s_i P^i, \quad (20)$$

with $i, n \in \mathbb{Z}^+$, n the order of the approximated polynomial, and $d_i, s_i \in \mathfrak{R}$ are the polynomial coefficients correspondingly.

The method for the case of estimating the damping and spring coefficients could be simplified by ignoring the inertial term $M\ddot{x}$ and considering (2) as a first order system, resulting in a rough initial estimation of the D, S parameters (Reynolds et al., 2003). Thus, Eq. (2) becomes:

$$D\dot{x} + Sx = F_{ce} - F_L, \quad (21)$$

For step changes in pressure and $F_{ce} > F_L$, the solution of (21) yields to:

$$x(t) = \frac{F_{ce} - F_L}{S} (1 - e^{-(S/D)t}) = \Delta x_{\max} (1 - e^{-(S/D)t}) \quad (22)$$

where $\Delta x_{\max}$ is the maximum contraction at steady state. To provide independent estimates of the model elements as functions of the pressure P , two separate inflation and deflation studies were conducted.

During the estimation related experiments, the proportional pressure regulator was disconnected from the pneumatic circuit. Instead of utilizing a regulator and inserting its dynamic characteristics into the PAM model approximation, the inflation and deflation experiments were performed by connecting the test-PAM directly to the pneumatic system's compressor. By utilizing a pneumatic switch as a step input function, the excitation of the test-PAM was performed via a pre-adjusted circuit to the respective test-pressure. Subsequently, via bypass of the regulator's dynamic effect, the PAM's response data was kept as unaffected by external dynamic influencing, as possible. Similar approaches to the technique described above have also been utilized in (Chou & Hannaford, 1996), (Reynolds et al., 2003) and (Tondur & Lopez, 2000).

Furthermore, the external effects concerning differences on how fast the air can get in and out of the muscle have been brought to a minimum. Specifically, during all the presented experimental trials, fast exhaust outlets were being placed directly after the pneumatic connection of the test-PAM, so as to allow the fastest possible air exhaustion.

For the inflation experiments, successive PAM contractions were conducted, with pressures between 1 and 6 bar (100–600 kPa) by 0.5 bar increments. The force coefficients F_{ce} at each pressure were



Fig. 5. PAM experimental setup.

estimated from (1) for $x = \Delta x_{\max}$, while the spring element S was estimated at each state by the following equation:

$$S = \frac{F_{ce} - F_L}{\Delta x_{\max}} \quad (23)$$

From (22), the time constant of the displacement response is equal to:

$$\tau = \frac{D}{S} \quad (24)$$

Based on these displacement responses the corresponding time constants were estimated and the damping element D was calculated at each state using (24).

For the deflation experiments, the same overall procedure was followed, but in this case the PAM performed successive extensions from initial pressure of 6 bar and final-state pressure of 1–6 bar by 0.5 bar decrements, in order to measure the respective maximum displacement $\Delta x_{\max}$ and, subsequently, compute the spring coefficients S from (23).

Additional, relaxation studies were conducted at which the PAM was subjected to successive deflations, with initial-state pressures between 1 and 6 bar (100–600 kPa) by 0.25 bar increments and zero final-state pressure. Based on these displacement responses the corresponding time constants were estimated and the damping element D was calculated at each state from (24) by utilizing the spring coefficients S extracted from the deflation experiments.

As it has been already mentioned, ignoring the inertial term $M\ddot{x}$ and considering (2) as a first order system results in a rough initial estimation of the D , S parameters. In order to further improve that estimation, extended manual fine-tuning was performed on the simulated nonlinear PAM system of (2), concerning inflation/deflation operating states.

The related bibliography regarding the phenomenological modeling approach of the PAM have included the utilization of 1st order polynomial fitting for the phenomenological coefficients S and D of a smaller in dimensions PAM (Reynolds et al., 2003) and a first order transfer function approximation of the static behavior of the PAM (Serres, Reynolds, Phillips, Rogers, & Repperger, 2010). For the fairly large in length and diameter PAM utilized in the presented experimental studies, the higher the order of the polynomial fitting of the aforementioned coefficients, the more accurate is the overall model approximation (Andrikopoulos et al., 2012a, 2012b). In addition, by taking into account the fact that the computational burden produced via the utilization of high-order polynomial fitting has decreased substantially in modern computerized systems, 4th order polynomials for the fitting of the spring and damping elements have been selected as an optimal trade-off between model approximation effectiveness and computational burden, since polynomial orders higher than this value do not significantly increase the effectiveness of the modeling process, and thus the adopted process cannot be considered as highly sensitive to the polynomial order selection.

Utilization of the aforementioned 4th-order polynomial fitting of the estimated S and D parameter values for inflation and deflation operating states, results in the respective polynomials displayed in (25).

$$\begin{aligned} S_{\text{inflation}}(P) &= 117.72P^4 - 1942.7P^3 + 11621P^2 - 29179P + 38535 \\ S_{\text{deflation}}(P) &= 32.77P^4 - 545.97P^3 + 3456.3P^2 - 8560.6P + 16977 \\ D_{\text{inflation}}(P) &= 79.4P^4 - 1325P^3 + 8183.7P^2 - 22409P + 28481 \\ D_{\text{deflation}}(P) &= -15.74P^4 + 230.55P^3 - 1016.5P^2 + 2602.7P + 269.77 \end{aligned} \quad (25)$$

Selecting the number of the utilized PWA models, denoted as E , is a process that requires a trade-off among the desired accuracy in the positioning task and the overall complexity of the resulting sMPC scheme. In general small E results in simpler controller

Table 1

PAM PWL approximation ($E=6$) characteristics for inflation and deflation operating states.

r	j	Operating pressure [bar]	Operating regions [m]	S	D	τ [s]
Inflation	1	1 $P \in [0, 1]$	$x \in [0, 0.0406]$	19400	13192	0.680
		2 $P \in [1, 2]$		15633	9726	0.622
		3 $P \in [2, 3]$	$x \in [0.0692, 0.0795]$	12883	6344	0.492
		4 $P \in [3, 4]$	$x \in [0.0795, 0.0851]$	13017	5384	0.414
		5 $P \in [4, 5]$	$x \in [0.0851, 0.0884]$	13633	5114	0.375
		6 $P \in [5, 6]$	$x \in [0.0884, 0.0915]$	14300	5195	0.362
Deflation	-1	1 $P \in [0, 1]$	$x \in [0, 0.0547]$	11200	1904	0.170
		2 $P \in [1, 2]$	$x \in [0.0547, 0.0758]$	10483	2635	0.251
		3 $P \in [2, 3]$	$x \in [0.0758, 0.0848]$	9823	3274	0.333
		4 $P \in [3, 4]$	$x \in [0.0848, 0.0874]$	10957	4586	0.419
		5 $P \in [4, 5]$	$x \in [0.0874, 0.0905]$	12183	6053	0.497
		6 $P \in [5, 6]$	$x \in [0.0905, 0.0915]$	13633	7722	0.566

architecture (small number of regions) with a small positioning accuracy, while a large E results in a controller with an increased complexity (large number of regions) and a high accuracy.

Based on the previous analysis and depending on the application requirements and existing computational capabilities, an iterative procedure can be adopted for tuning E . As the presented problem in this article concerns the fundamental modeling and control of a PAM, after the proper selection of E , this value will remain fixed in case of integrating the PAM component in a more complex robotic or automation structure.

Thus, having estimated the relations of $S(P)$ and $D(P)$, under the assumption of a constant L , the total number of PWA subsystems has been properly selected as $E=6$. Smaller number of linearization points worsens the modeling accuracy and controller's efficacy, while a larger number of regions results in an increased complexity with a slight increase in performance. Thus, for an optimal trade-off between complexity and performance, the following experimental evaluation of the sMPC scheme has been conducted with a modeling approximation of 6 linearization points.

In the case where $E=6$, operational regions' boundary elements have been selected as the steady state values, defined in (23) as $\Delta x_{\max}$, of the PAM system open loop response for pressures from 0 to 6 bar, in intervals of 1 bar; whereas their respective linearization points have been selected as the steady state values $\Delta x_{\max}$ of the PAM system open loop response for the middle-point pressures P^{op} from 0.5 to 5.5 bar, with 1 bar intervals.

Based on the assumption of linear relationships of $S(P)$ and $D(P)$ inside every operating region, the aforementioned coefficient values, as well as the operating parameters and their respective regions, are presented in Table 1. The PAM utilized in these experiments, mainly due to its fairly large dimensions, is characterized by inherently large settling times in the order of a second. This manufacturing fact has being elaborated further by the additional time constant data extracted from (24) and are also presented in Table 1.

Each linearized PWA subsystem is valid in only one of these regions. Therefore the constraints of x_j in the j th operating region are the corresponding boundaries of the region.

4.3. Control algorithm tuning

Throughout the tuning and the following evaluation experimental studies, the total displacement is constrained at $0 \leq x \leq 0.0915$ (m), no constraint is posed on the state velocity $\dot{x}$, the affine state $f_{j,r}^{au}$ of the j th operating region is kept constant at the respective value utilized in (14), the control effort is constrained at

$-100 \leq \delta P \leq 100$ (kPa), and no constraint is posed on the increment rate of u .

The goal of the proposed control scheme is to solve the set-point problem, which drives all outputs towards the desired ones (set-point regulation). The selection of the MPC's cost function weighting parameters U_j , V_j and W_j is based on the goal of adjusting the respective three terms of the objective function (17) to a comparable point and therefore achieve an optimal trade-off between reference tracking and control speed. For that reason, the difference in input units and output units, which are constrained in the order of 10^6 Pa and 10^{-3} m, respectively, is being taken into consideration and is being reflected in the selection of the function parameters, thus, resulting in different numerical ranges. After iterative search and extensive experimentation the aforementioned parameters were properly set to $U_j=0.5$, $V_j=20$ and $W_j=10^6 \mathbf{I}_3$ for $j=1, \dots, 6$.

The online nature of the MPC algorithm, regarding the QP optimization cost solution, as well as the PWA model approximation, which creates the need for a switching structure of the multiple MP-controllers, lead to an increase in the overall computational burden as a long prediction horizon is usually needed to guarantee feasibility (Mayne et al., 2000). In the sequel, an experimental study regarding the selection of the prediction N_P and control N_C horizon and their effect in the control structure's complexity and performance is being presented, in order to analyse and investigate the optimal trade-off between the two aforementioned control attributes.

The ad hoc parameters' selection for the prediction N_P and control N_C horizon was finalized after performing multiple experimental responses for various pairs of N_P , N_C parameters in order to pinpoint the values for the optimal set-point regulation of the PAM. It should be noted that controllers with $N_P < N_C$ are truncated due to unfeasibility and, in this cases, the control algorithm is being decreased, thus, forcing an equality $N_P = N_C$. The aforementioned cases have been properly outlined in Figs. 6 and 7.

The mean absolute steady state error data extracted from the PAM responses via step reference signals of random amplitudes is depicted in Fig. 6, while the truncated control cases are being outlined via red dotted lines. The aforementioned data showed that for the PAM system the response error reaches a minimum of approximately 0.14% for $N_P=5$ and $N_C=5$ and a maximum of approximately 0.8% for $N_P=21$ and $N_C=9$.

In Fig. 7, the mean execution times of the implemented control algorithm are being depicted, while the truncated control cases are being outlined via red dotted lines. These experimental data concern the corresponding same test cases via the pairs of N_P , N_C

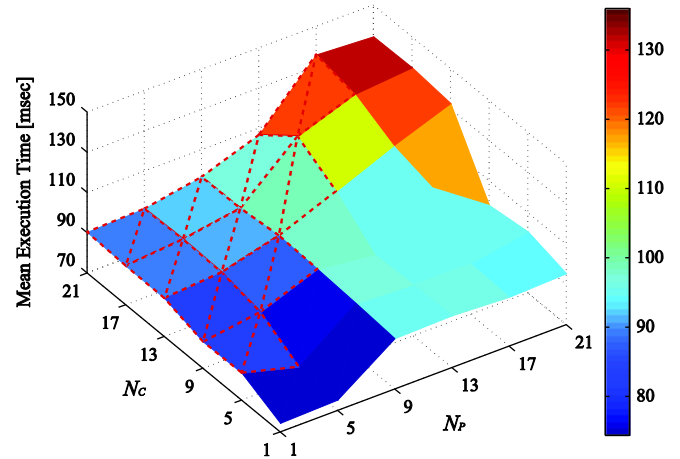


Fig. 7. Mean execution time (ms) of the control scheme implementation with respect to the prediction horizon N_P and the control horizon N_C .

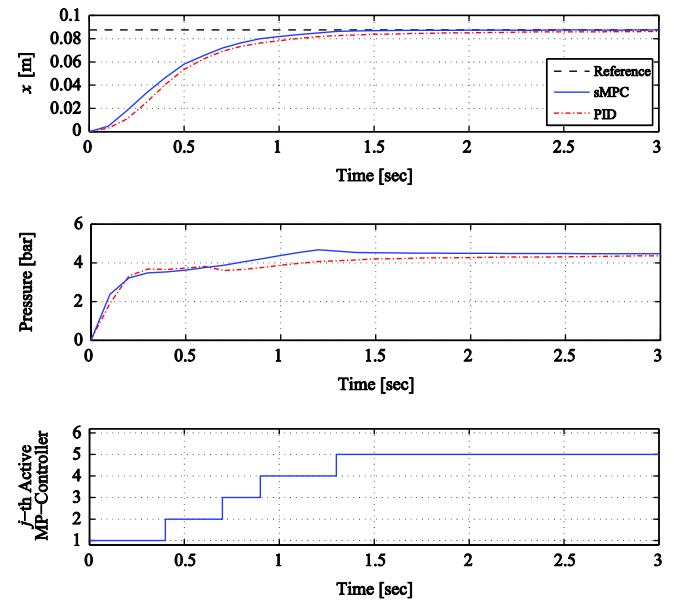


Fig. 8. The experimental PAM system response for initial states of $\mathbf{x}_{\text{initial}}^{\text{au}} = [0, 0, 0]$ and desired states of $\mathbf{x}_{\text{ref}}^{\text{au}} = [0.0876, 0, f_5^{\text{au}}]$.

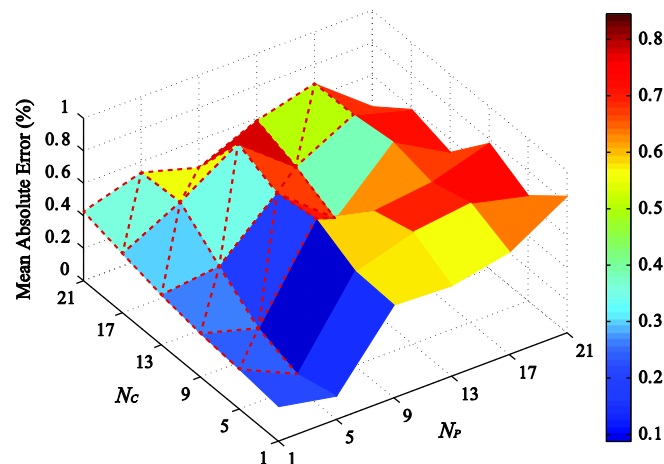


Fig. 6. Mean absolute error (%) with respect to the prediction horizon N_P and the control horizon N_C .

parameters presented in Fig. 6. In overall, the computational burden remains larger than 70 ms for all pairs of prediction and control horizon due to the online minimization of an optimal control cost. In addition, the mean execution time increases monotonically with relation to N_P and N_C , giving a minimum execution time of approximately 75 ms for $N_P=5$ and $N_C=5$ and a maximum of approximately 140 ms for $N_P=21$ and $N_C=21$.

By taking all data into consideration, in order to achieve an optimal trade-off between control complexity and optimal performance, the parameters involved in the QP optimization cost (17) were fine-tuned to $N_P^j=5$, $N_C^j=5$. It should be noted that the algorithm's execution time plays a major role in the selection of the sampling period T_s , which in this case was adjusted to $T_s=100$ ms.

4.4. sMPC experimental validation

The experimental data presented below are being accompanied by responses acquired via PID control of the PAM setup. The purpose is to compare the response attributes and characteristics of the proposed control scheme with those of a conventional and

widely-used control strategy. The PID control parameters utilized in these experiments have been fine-tuned after extensive trial and error sequences.

The comparison with the PID control scheme has been selected since the PID controller is the one that have been mostly being utilized in the related literature of PAMs, especially with PAMs in real life applications. The PID scheme as also all the other mentioned control schemes in the area of PAMs have not the ability to handle: (a) constraints on the states and on the control signals, demands that are always found in real life applications and (b) multiple piecewise system representations that could increase the PAM modeling accuracy. As it has been indicated, in cases

where a very good performance in positioning is needed the proposed scheme is able to deliver very good accuracy and fast responses and it should be considered a functional and experimentally proven additional control approach to the related existing ones.

The experimental PAM system response for initial states of $\mathbf{x}_{\text{initial}}^{\text{au}} = [0, 0, 0]$ and desired states of $\mathbf{x}_{\text{ref}}^{\text{au}} = [0.0876, 0, f_{5,1}^{\text{au}}]$, is depicted in Fig. 8, along with the corresponding overall control actuation $P = P_0 + \delta P$ and the response of the current active j MP-controller. The PID response, which is also being displayed in the same figure, has the respective gain parameters tuned to $K_p = 124.2$, $K_i = 95$, $K_d = 3.9$.

Regarding the aforementioned inflation response, the sMPC implementation is being characterized by smooth and rapid convergence, absence of state x oscillations and mean steady state error of less than 0.14%, resulting in fast and high-accuracy set-point regulation. The respective PID response features larger settling time and mean steady state error of approximately 1.8%, showing the superiority of the proposed sMPC scheme in regulating the position of the PAM.

In addition, the experimental PAM system response for initial states of $\mathbf{x}_{\text{initial}}^{\text{au}} = [0, 0.0911, f_{6,1}^{\text{au}}]$ and desired states of $\mathbf{x}_{\text{ref}}^{\text{au}} = [0.082, 0, f_{4,-1}^{\text{au}}]$, is depicted in Fig. 9, along with the corresponding overall control actuation $P = P_0 + \delta P$ and the response of the current active j MP-controller.

It should be noted that Fig. 9 juxtaposes the displacement response of the proposed dual-switch MPC scheme, denoted as $x_{\text{dual-switch}}$ to emphasize the double switching between inflation and deflation operating states and their PWA subsystems, with the displacement response of the scheme's function incorporating only the switching between inflation PWA subsystems, denoted as $x_{\text{single-switch}}$. For the two aforementioned responses, only the dual-switch approach manages to successfully achieve set-point regulation, thus, showing the necessity of incorporating the alteration of PAM dynamics during inflation and deflation operating states into the modeling approaches and control algorithms. The 'dual-switch' deflation response achieves rapid convergence and absence of state fluctuations, resulting in mean steady state

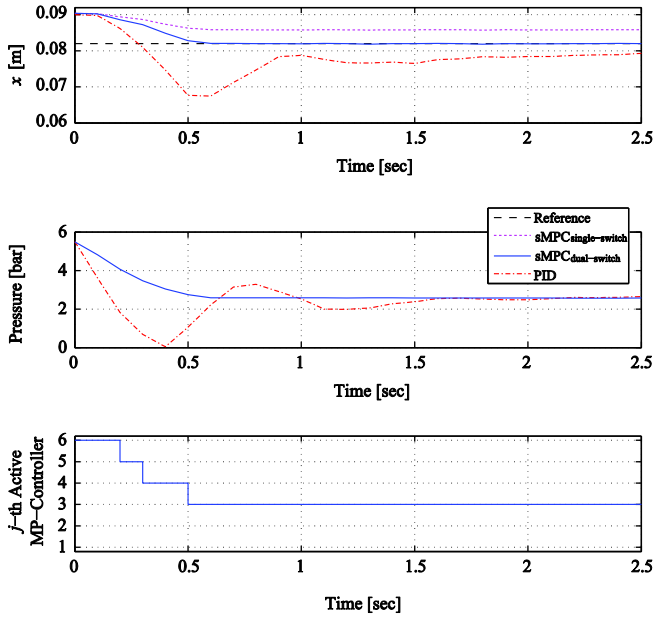


Fig. 9. The experimental PAM system response for initial states of $\mathbf{x}_{\text{initial}}^{\text{au}} = [0, 0, 0]$ and desired states of $\mathbf{x}_{\text{ref}}^{\text{au}} = [0.0856, 0, f_5^{\text{au}}]$.

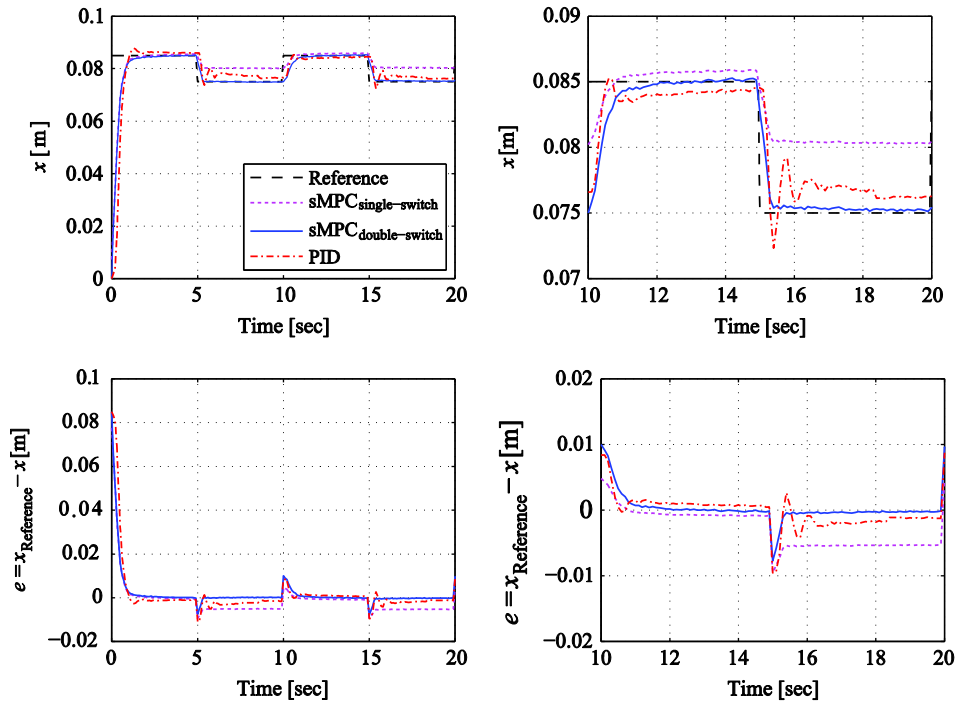


Fig. 10. The experimental PAM system response for initial states of $\mathbf{x}_{\text{initial}}^{\text{au}} = [0, 0, 0]$ and pulse reference signal.

error of 0.015%. The respective PID controller, with gain parameters tuned to $K_p=450.2$, $K_i=65.3$, $K_d=2.5$, shows slow convergence during deflation and large mean steady state error, while any further attempt to increase the convergence speed causes intense transient phenomena.

In the sequel, the experimental responses of the PAM's displacement x , an enlarged portion of the response for ameliorated visualization and the respective error signal $e = x_{ref} - x$, under a pulse reference signal ranging between 0.075 and 0.085 mm, and a sinusoidal reference signal ranging between 0.05 and 0.09 mm at 0.05 Hz, are presented in Figs. 10 and 12, respectively.

As it can be observed in Fig. 10, in the beginning of the experimentation the PAM is in a state that it has not been inflated

and thus the tip of the PAM is relying on the zero (initial position). A reference command of a displacement for the tip of the PAM to the 0.08 m position will initially and instantaneously create a big error, difference between the current and the requested positioning. As the time evolves and the PAM is appropriately controlled to reach the final reference position, this indicating positioning error in Fig. 10 is converging to zero.

For the obtained translational movement of the PAM, the respective responses of the velocity $\dot{x}$, the overall control effort and the currently active j MP-controller are displayed in Figs. 11 and 13 for both of the examined test cases respectively. All the dual-switch responses are characterized by smooth control effort transition, absence of intense overshoot phenomena, even during

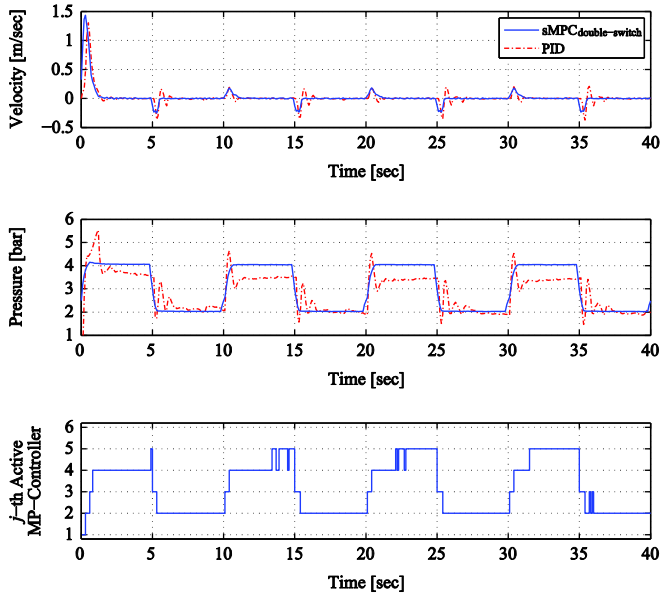


Fig. 11. The overall control effort $P = P_0 + \delta P$, operating state r and j th active MP-controller for the reference signal displayed in Fig. 10.

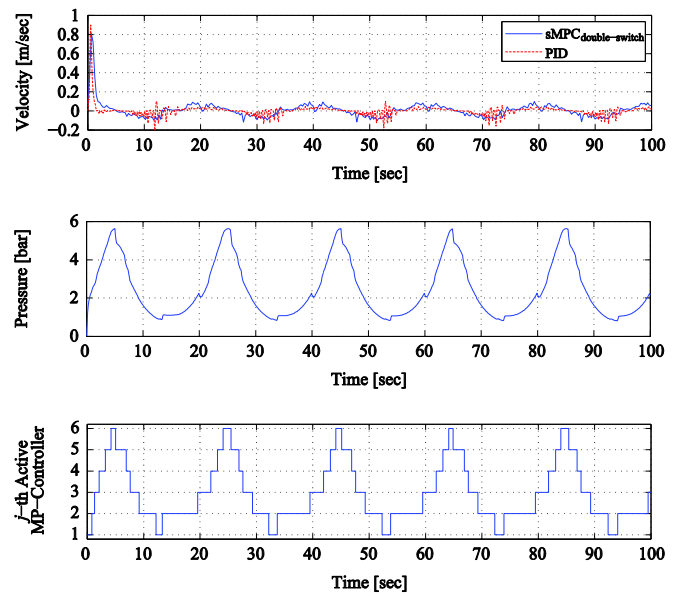


Fig. 13. The overall control effort $P = P_0 + \delta P$, operating state r and j th active MP-controller for the reference signal displayed in Fig. 12.

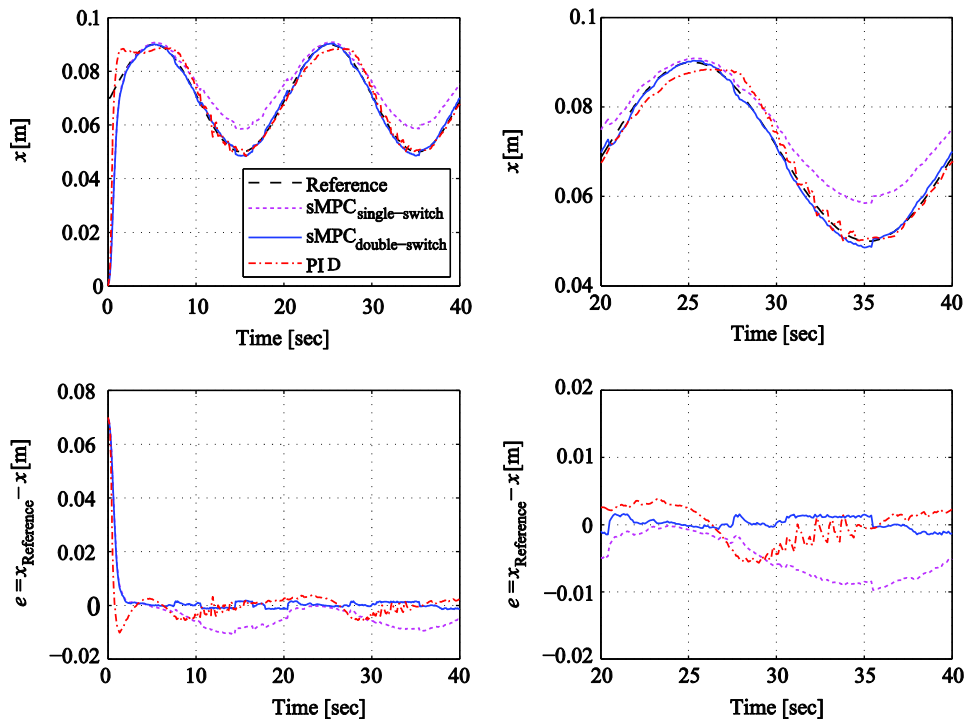


Fig. 12. The experimental PAM system response for initial states of $x_{initial}^{du} = [0, 0, 0]$ and sinusoidal reference signal.

rapid switching between neighboring regions, which results in very small x state variations from the respective sinusoidal and pulse reference signals. The single-switch approach fails to track the deflation segments of the reference signals, thus, evaluating the incorporation of the dynamic alterations between inflation and deflation operating states, as imperative for modeling and optimal control of the PAM.

Comparison to the respective PID responses, with gain parameters tuned to $K_p=144.1$, $K_i=90$, $K_d=3.5$ for the pulse reference signal and $K_p=410.5$, $K_i=72$, $K_d=3$ for the sinusoidal reference signal, shows the advanced efficacy of the proposed sMPC scheme in achieving fast, smooth and accurate responses comparable to the fluidic PAM's inherently large settling times.

Overall, from the obtained experimental responses, the proposed dual scheme, based on the model predictive control principle, has a well performance independently of the reference signal type. The steady state is characterized by a smooth transition, a fast response, and without the existence of overshoots, during both inflation and deflation modes of operation.

5. Conclusions

In this article, a switching Model Predictive Control (sMPC) scheme for the position control of a pneumatic artificial muscle (PAM) was presented. The control scheme was based on constrained linear and PieceWise Affine (PWA) system model approximation. Experimental studies were utilized to investigate and evaluate the efficacy of the suggested controller in the positioning problem of a PAM. The proposed sMPC scheme resulted in an optimal control of each operating region and ensured the control effort's smooth transition as the system switched between neighboring regions. As it has been presented, by retaining relatively small prediction and control horizons and corresponding small order for the polynomial representation of the spring element, a significantly high precision and fast response for the PAM can be achieved. In the case of a different PAM device the proposed modeling and control scheme can be directly applied, while the user needs to perform a small set of parameter fine tuning based on the strategy and the approaches presented. Overall, the aim of this article was to design, fine tune and experimentally evaluate a switching MPC scheme based on the novel proposed PAM modeling, especially for applications where maximum positioning accuracy, fast responses with low overshoot and multiple constraints in the PAM operation should be taken under consideration. Except from the superiority of the proposed scheme, when compared with the existing techniques and their ability to take under consideration the constraints, in a general engineering problem the selection of the control structure should be always performed based on the application demands and the computational burden that the existing computational system could handle.

Appendix A. List of variables and utilized symbols

$F_{ce} \in \mathbb{R}$	contractile force of the PAM
$l \in \mathbb{R}^+$	nominal length of the PAM, with an initial value of l_0
$r \in \mathbb{R}^+$	bladder radius of the PAM, with an initial value of r_0
$\theta \in \mathbb{R}$	angle between the helical fiber element length and the longitudinal axis of the PAM, with an initial value of θ_0
$M \in \mathbb{R}^+$	mass of the PAM
$D \in \mathbb{R}^+$	damping element of the PAM's phenomenological representation
$S \in \mathbb{R}^+$	spring element of the PAM's phenomenological representation

$F_L \in \mathbb{R}$	external load of the PAM's phenomenological representation
$x \in \mathbb{R}^+$	displacement of the PAM
$P \in \mathbb{R}^+$	pressure applied inside the PAM
$()^{op}$	notation for denoting the operating points
j	symbol for denoting the variables related to the j th operating point
$E \in \mathbb{Z}^+$	maximum number of switched systems
$\delta x \in x$	small perturbation around x
$\delta P \in P$	small perturbation around P
$X \in \mathbb{Z}^2$	state operating region
$P \in \mathbb{R}$	input operating region
$\mathbf{x} \in X$	state vector of the PAM's state space representation
$\mathbf{y} \in P$	output vector of the PAM's state space representation
$\mathbf{A} \in \mathbb{R}^{2 \times 2}$	state matrix of the j th sub-system
$\mathbf{B} \in \mathbb{R}^{2 \times 1}$	input matrix of the j th sub-system
$\mathbf{C} \in \mathbb{R}^{2 \times 2}$	output matrix of the j th sub-system
$\mathbf{D} \in \mathbb{R}^{2 \times 1}$	feed-through matrix of the j th sub-system
$\mathbf{f} \in \mathbb{R}^{2 \times 1}$	affine term of the j th sub-system
Σ	polytope that contains all the switching systems
$\text{Co}\{\}$	notation for denoting the convex hull of the set Σ
$T_s \in \mathbb{R}^+$	sampling period
$k \in \mathbb{Z}^+$	symbol for denoting the k th sample
Ω	polyhedral partition of the state-input space
$()^*$	notation for denoting the discrete equivalents of the PWA matrices
$()_{au}$	notation for denoting the augmented matrices of the PWA sub-systems where the affine term is included in the state vector
$()_i$	notation for denoting the i th component's prediction
$()_l$	notation for denoting the l th component of a vector
$u \in \mathbb{R}$	control effort
$\Delta u \in \mathbb{R}$	control effort increments
$y \in \mathbb{R}$	control outputs
$\varepsilon \in \mathbb{R}^+$	relaxing slack variable
$k+ilk$	expression denoting the predicted value for time $k+i$ based on information available at the k -time instant
$N_p \in \mathbb{Z}^+$	prediction horizon of the j th MP-controller
$N_c \in \mathbb{Z}^+$	control horizon of the j th MP-controller
$U \in \mathbb{R}^+$	positive semi-definite penalty weight on the input u of the j th MP-controller
$V \in \mathbb{R}^+$	positive semi-definite penalty weight on the input increments Δu of the j th MP-controller
$W \in \mathbb{R}^+$	positive semi-definite penalty weight on the output y of the j th MP-controller
$P_0 \in \mathbb{R}^+$	control effort of the feed-forward controller
$x_{ref} \in \mathbb{R}^+$	reference position
$r \in \mathbb{Z}$	Schmitt trigger-based signal denoting the operating state of the PAM
$g \in \mathbb{R}^+$	empirically fine-tuned constant posed on the switch signal r
$u_{target} \in \mathbb{R}^+$	input set-point value
ρ_e	penalty weight on the slack variable ε
$(k k)^*$	expression denoting the first element of the optimal sequence

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